

ECHS MATHEMATICS

IB Diploma Programme - Mathematics: Applications and Interpretation SL

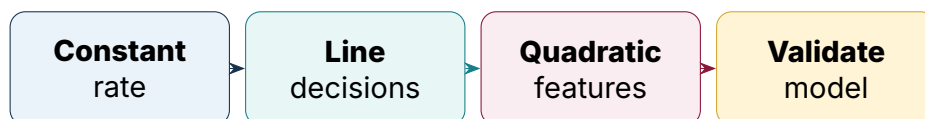
2.2

Linear and Quadratic Models

Full Notes - Rate - Intersections - Roots - Vertex - Modelling - TI-84 Evidence - Validation

Central question. When is constant change enough, and when does curvature produce a better model? A defensible model links structure, parameters, domain, technology evidence, and contextual judgement.

model → solve → verify → interpret → evaluate



Name: _____ Class: _____

Teacher: Mohammad Abu-Ghuwaleh

Version: Student Full Notes

Learning route and reference map

structure → equation → calculator evidence → contextual decision

Learning intentions

By the end of Lesson 2.2, you should be able to:

1. construct and interpret linear models from rates, points, tables, and contexts;
2. use parallel, perpendicular, and intersection reasoning in decisions;
3. connect standard, factored, and vertex forms of a quadratic to graph features;
4. construct quadratic models from roots, a vertex, three points, or a context;
5. solve linear/quadratic equations and intersections exactly when possible;
6. use TI-84 Zero, Maximum/Minimum, Intersect, and TABLE transparently;
7. use differences and residuals to compare model families;
8. state domains, assumptions, limitations, and extrapolation risk.

Core reference

Linear model

$y = mx + c$, where m is constant rate and $c = f(0)$ is the initial/-fixed value.

Gradient

$m = \frac{y_2 - y_1}{x_2 - x_1}$, including units.

Parallel

Equal gradients for distinct non-vertical lines.

Perpendicular

$m_1 m_2 = -1$ for non-vertical lines.

Quadratic standard form

$q(x) = ax^2 + bx + c$, $a \neq 0$.

Factored form

$q(x) = a(x - r_1)(x - r_2)$ reveals roots.

Vertex form

$q(x) = a(x - h)^2 + k$ reveals vertex (h, k) .

Axis of symmetry

$x = -\frac{b}{2a} = \frac{r_1 + r_2}{2}$ when roots are real and distinct.

Discriminant

$b^2 - 4ac$ classifies the number of real roots.

Residual

$r = y_{\text{observed}} - y_{\text{predicted}}$.

IB communication tip

Manual mathematics stays visible. Use a calculator where it genuinely adds evidence, then independently verify the output and interpret it in the stated domain.

2.2A - Linear Models and Constant Rate

rate and initial value

A linear model has constant rate

For equal changes in the independent variable, a linear model gives equal changes in the dependent variable.

$$y = mx + c.$$

The gradient m has units of *output units per input unit*. The intercept $c = f(0)$ is an initial or fixed value when $x = 0$ has contextual meaning.

Worked Example 1

GDC NOT NEEDED

A delivery charge is 42 QAR for 5 km and 78 QAR for 17 km. Find and interpret the gradient.

WORKING / ANNOTATIONS

Your Turn 1

GDC NOT NEEDED

A tank contains 68 L after 2 minutes and 48 L after 7 minutes. Find the gradient and interpret its sign and units.

WORKING / ANNOTATIONS

Build a model from context

A gym charges a 150 QAR joining fee and 45 QAR per month:

$$C(n) = 45n + 150.$$

After 8 months, $C(8) = 510$ QAR. If n counts completed months, a natural domain is $n \in \{0, 1, 2, \dots\}$ while the pricing policy remains unchanged.

Line through two points

First calculate the gradient, then use point-gradient form:

$$y - y_1 = m(x - x_1).$$

Through (4, 19) and (10, 43):

$$m = 4, \quad y - 19 = 4(x - 4) \Rightarrow \boxed{y = 4x + 3}.$$

First differences

For equally spaced inputs, constant first differences support a linear model. If outputs are

7, 11, 15, 19, 23,

the constant difference is 4, so for inputs 0, 1, 2, 3, 4 the model is $y = 4x + 7$.

Checkpoint - Linear models

1. Find the line through $(-2, 9)$ and $(4, -3)$.
2. Explain what a gradient means in a cost-versus-distance model.
3. Explain why constant first differences require equal input spacing.
4. State a realistic domain when the input counts students in a room of capacity 28.

WORKING / ANNOTATIONS

2.2B - Line Relationships and Decisions

gradients, intersections, and constraints

Parallel and perpendicular

For non-vertical lines:

$$\text{parallel: } m_1 = m_2, \quad \text{perpendicular: } m_1 m_2 = -1.$$

A horizontal line and a vertical line form a special perpendicular pair.

Worked Example 2**GDC NOT NEEDED**

Find the line perpendicular to $y = \frac{1}{2}x + 7$ through $(4, -1)$.

WORKING / ANNOTATIONS**Your Turn 2****GDC NOT NEEDED**

Find the line parallel to $3x - 2y = 8$ through $(-2, 5)$.

WORKING / ANNOTATIONS**Intersections are simultaneous conditions**

To find where two models have equal output, solve $f(x) = g(x)$ and then obtain the common output. Always report the coordinate pair when the question asks for an intersection.

Worked Example 3**GDC NOT NEEDED**

Plan P costs $35 + 1.8x$ QAR and Plan Q costs $11 + 2.6x$ QAR. Find and interpret the break-even point.

WORKING / ANNOTATIONS**Common trap**

A correctly solved intersection can still be irrelevant if it lies outside the contextual domain. Apply domain, units, and discreteness after solving.

2.2C - Quadratic Structure and Exact Features

roots, intercepts, symmetry, and extrema

Three equivalent forms

The same parabola may be written in three useful forms:

$$x^2 - 6x + 5 = (x - 1)(x - 5) = (x - 3)^2 - 4.$$

- Standard form shows a, b, c and y -intercept c .
- Factored form shows roots 1 and 5.
- Vertex form shows vertex $(3, -4)$ and axis $x = 3$.

Discriminant and root structure

For $ax^2 + bx + c = 0$:

$$\Delta = b^2 - 4ac.$$

$\Delta > 0$: two distinct real roots; $\Delta = 0$: one repeated real root; $\Delta < 0$: no real roots. A repeated quadratic root touches the x -axis and turns.

Worked Example 4

GDC NOT NEEDED

For $q(x) = -2x^2 + 12x + 14$, determine the roots, y -intercept, axis, and maximum.

WORKING / ANNOTATIONS

Your Turn 3

GDC NOT NEEDED

For $p(x) = x^2 - 8x + 12 = (x - 2)(x - 6)$, state roots, axis, vertex classification, and y -intercept.

WORKING / ANNOTATIONS

Restricted domain changes range

For $q(x) = 2(x - 3)^2 - 4$ unrestricted, the range is $y \geq -4$. On $0 \leq x \leq 5$, compare the vertex and endpoints:

$$q(3) = -4, \quad q(0) = 14, \quad q(5) = 4.$$

Hence $-4 \leq y \leq 14$.

Checkpoint - Quadratic structure

1. For $y = 3(x - 2)(x + 4)$, state roots and axis.
2. For $y = -2(x + 1)^2 + 9$, state vertex and maximum.
3. Find the y -intercept of $2x^2 - 5x + 7$.
4. Explain why two roots do not determine a unique quadratic.

WORKING / ANNOTATIONS

2.2D - Constructing Quadratic Models

choose the form that matches the information

Worked Example 5

GDC NOT NEEDED

An arch meets the ground at $x = 0$ and $x = 12$ and reaches height 9 at $x = 6$. Find a model and contextual domain.

WORKING / ANNOTATIONS

Your Turn 4

GDC NOT NEEDED

A quadratic has roots -2 and 5 and passes through $(0, -20)$. Construct and expand the model.

WORKING / ANNOTATIONS

Vertex plus one point

If the vertex is $(4, -3)$ and the graph passes through $(6, 5)$, use

$$y = a(x - 4)^2 - 3.$$

Then $5 = 4a - 3$, so $a = 2$ and $y = 2(x - 4)^2 - 3$.

Three points determine the coefficients

For points $(0, 4)$, $(2, 10)$, $(5, 49)$, substitute into $q(x) = ax^2 + bx + c$. Solving the resulting three-equation system gives

$$q(x) = 2x^2 - x + 4.$$

A calculator regression may check the result, but the conditions should still be written mathematically.

Projectile model

$$h(t) = -4.9t^2 + 18t + 1.5.$$

$h(0) = 1.5$ m is launch height; the vertex gives maximum height/time; the positive zero gives landing time. The negative zero is mathematically valid but rejected by $t \geq 0$.

Worked Example 6

GDC NOT NEEDED

Demand is $n = 800 - 20p$, where p is ticket price. Construct the revenue model and find its maximum.

WORKING / ANNOTATIONS

Worked Example 7

GDC NOT NEEDED

A rectangular enclosure uses 100 m of fencing for three sides beside a river. Find the maximum area.

WORKING / ANNOTATIONS

2.2E - Solving and TI-84 Evidence

exact structure first, technology where useful

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $3x^2 + x - 5 = 0$:

$$x = \frac{-1 \pm \sqrt{61}}{6} \approx 1.13504, -1.46837,$$

so to 3 significant figures, $x \approx 1.14, -1.47$.

Technology evidence chain

GDC EXPECTED State the model; record Y_1, Y_2 or expression; use a suitable domain/window; report every relevant output; verify independently; interpret units, domain, and precision.

TI-84 Workflow A - Zero

GDC EXPECTED

For $h(t) = -4.9t^2 + 18t + 1.5$, find the positive zero. **1.** $Y=$; enter $-4.9X^2 + 18X + 1.5$ in Y_1 .

2. Use a contextual window such as $0 \leq X \leq 4.2, -2 \leq Y \leq 20$.

3. GRAPH; 2nd TRACE (CALC) $\rightarrow$ 2:zero.

4. Set Left Bound, Right Bound, and Guess around the positive crossing.

OUTPUT. $t \approx 3.7550$.

VERIFY. Substitution gives a residual near zero; reject the negative root $t \approx -0.0815$ because $t < 0$.

TI-84 Workflow B - Maximum / Minimum

GDC EXPECTED

For $R(p) = -20p^2 + 800p, 0 \leq p \leq 40$, graph a window containing the vertex. Use 2nd TRACE $\rightarrow$ 4:maximum. The output is (20, 8000). Verify using $p = -b/(2a) = 20$ and substitution.

TI-84 Workflow C - Intersect**GDC EXPECTED**

Find all intersections of

$$y = 2x + 7, \quad y = -x^2 + 10x + 3.$$

Enter both functions, graph a window showing both crossings, then use 2nd TRACE → 5:intersect near *each* crossing.

$$(0.535898, 8.071797), \quad (7.464102, 21.928203).$$

Exact algebra gives $x = 4 \pm 2\sqrt{3}$ and $y = 15 \pm 4\sqrt{3}$.**TI-84 Workflow D - TABLE for whole-number thresholds****GDC EXPECTED**For $C(n) = 0.5n^2 + 8n + 20$, find the first whole n with $C(n) \geq 160$. The continuous crossing is about 10.55. Set TblStart=9 and $\Delta\text{Tbl}=1$:

$$C(10) = 150 < 160, \quad C(11) = 168.5 \geq 160.$$

Therefore $\boxed{n = 11}$.**Common trap**

A default graph window can hide a root or intersection. "Not visible" is not evidence of non-existence. Use domain reasoning, a table, or algebra to choose a defensible window.

2.2F - Model Choice, Validation, and IB Synthesis

fit is evidence, not proof

Difference structure

For equally spaced inputs:

constant first differences $\Rightarrow$ linear evidence,constant second differences $\Rightarrow$ quadratic evidence.For outputs 5, 8, 13, 20 at $t = 0, 1, 2, 3$, first differences are 3, 5, 7 and second differences are 2, 2, giving $q(t) = t^2 + 2t + 5$.**Residuals**

$$r = y_{\text{observed}} - y_{\text{predicted}}.$$

Small, patternless residuals support the model. A systematic curve in residuals suggests a linear fit has missed curvature. One small residual never proves the whole model is suitable.

Interpolation versus extrapolation

Interpolation predicts inside the observed/validated interval and is usually lower risk. Extrapolation extends beyond evidence, where relationships, capacities, or policies may change.

Model evaluation checklist

Ask:

- Do the parameters have meaningful units and signs?
- Does the domain match physical or policy restrictions?
- Are residuals reasonably small and patternless?
- Which assumptions are being held constant?
- Which variables are omitted?
- Is the conclusion interpolation or extrapolation?

IB-style Core Tasks

calculate, verify, and interpret

IB Task A - Transport plans and break-even

GDC USEFUL

Two providers charge

$$A(d) = 180 + 3.2d, \quad B(d) = 60 + 5.6d,$$

where d is kilometres and cost is QAR.

- Interpret the gradient and intercept of each model. [4]
- Determine and interpret the break-even point. [4]
- Determine which plan is cheaper at 80 km and by how much. [2]
- State one assumption and one limitation. [4]

WORKING / ANNOTATIONS

IB Task B - Parabolic pedestrian bridge**GDC EXPECTED**

An arch meets the ground at $x = 0$ and $x = 18$ m. Its maximum height is 8 m at the centre.

- (a) Construct $h(x) = ax(x - 18)$. [4]
- (b) State a contextual domain and range. [3]
- (c) Find the height 3 m from the left support. [3]
- (d) A vehicle requires 5 m vertical clearance. Determine the interval of x for which clearance is at least 5 m. [5]

WORKING / ANNOTATIONS

IB Task C - Projectile model

GDC EXPECTED

For

$$h(t) = -4.9t^2 + 18t + 1.5,$$

where t is seconds and h is metres:

- (a) State the launch height. [2]
- (b) Determine maximum height and when it occurs. [5]
- (c) Use TI-84 Zero to determine when the ball reaches the ground and record relevant evidence. [5]
- (d) State one limitation. [4]

WORKING / ANNOTATIONS

Mastery Checklist and TI-84 Quick Reference

evidence before moving to broader models

Mastery checklist

I can independently:

- interpret gradient and intercept with units;
- construct linear equations and solve break-even decisions;
- identify quadratic roots, intercepts, axis, vertex, and extremum;
- move between standard, factored, and vertex forms;
- construct quadratic models from roots, vertices, points, and context;
- use exact algebra before rounding;
- use Zero, Maximum/Minimum, Intersect, and TABLE with reproducible calculator evidence;
- verify calculator output and apply domain/discreteness restrictions;
- use differences and residuals to justify model choice;
- evaluate assumptions, limitations, and extrapolation risk.

TI-84 quick reference

Zero

$Y = \rightarrow$ GRAPH $\rightarrow$ 2nd TRACE $\rightarrow$ 2:zero $\rightarrow$ bounds $\rightarrow$ Guess.

Maximum / Minimum

Graph the vertex; 2nd TRACE $\rightarrow$ 4:maximum or 3:minimum; bound the vertex.

Intersect

Enter Y_1, Y_2 ; 2nd TRACE $\rightarrow$ 5:intersect; repeat near every relevant crossing.

TABLE

2nd WINDOW (TBLSET) $\rightarrow$ set TblStart and Δ Tbl $\rightarrow$ 2nd GRAPH (TABLE); compare adjacent whole inputs.

IB communication tip

A high-quality final line answers the context: name the chosen variable value, units, relevant domain restriction, precision, and why any alternative mathematical solution is rejected.